A MULTI-MODAL FRAMEWORK FOR QUANTIFYING NARRATOR RELIABILITY IN 'ILM AR-RIJAL USING SEMANTIC SENTIMENT ANALYSIS AND GRAPH NEURAL NETWORKS

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DECLARATION

I hereby declare that the material in this thesis is my own except for quotations, equations, summaries and references, which have been duly acknowledged.

21 APRIL 2026

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"خَيْرُ النَّاسِ أَنْفَعُهُمْ لِلنَّاسِ"
"*The best of people are those who are most beneficial to others.*"
– Prophet Muhammad ﷺ

(At-Tabarani in Al-Mu'jam al-Awsat Hadith: 5787, graded *Sahih* by Al-Albani)

All praise and gratitude belong to Allah for giving me the strength, patience, and clarity to complete this proposal. This work is a small attempt to serve the tradition of 'Ilm ar-Rijal, and I pray that it benefits others in some way.

I would like to express my deepest thanks to my supervisor, Dr. Khalifa Chekima, for his guidance, patience, and honest feedback throughout this journey. He gave me the freedom to explore an ambitious idea while quietly making sure I didn't go off the rails.

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May Allah accept this small effort of mine, forgive my shortcomings, and make it beneficial to those who study His religion.

Alan Syahmi Bin Sanih @ Sani
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ABSTRACT

The authentication of Prophetic reports (Ḥadith) through the science of narrator criticism (*ʿIlm ar-Rijal/Jarḥ wa Taʿdīl*) is a cornerstone of Islamic scholarship, yet the digital era has outpaced manual verification methods. This project proposes a multi-modal framework that automates narrator reliability scoring by integrating three computational modalities: Arabic Named Entity Recognition (NER) using a fine-tuned AraBERT v2 model to extract narrator names, kunya, and nisba from classical biographical texts (Taqrīb al-Tahḏīb of Ibn Ḥajar al-Asqalāni and the Tahḏīb al-Kamāl of al-Mizzī) and *isnād* chains (AR-Sanad 280K); a Knowledge Graph (KG) constructed with NetworkX (~1,500 narrator nodes, verbal edges e.g. heard_from and reported_from, rich attributes including *tabaqa*, travel history, and classical *Jarḥ wa Taʿdīl* labels) to enable structural and temporal integrity checks; and semantic sentiment quantisation that maps qualitative terms (e.g., *Thiqa*, *Ṣadūq*, *Kaḏḏāb*) to continuous reliability weights (0.00–1.00) and applies edge-level risk modifiers (e.g., 0.80 for *tadlīs*). The final chain score is computed as the product of node weights and edge modifiers, then mapped to *Ṣaḥīḥ/Ḥasan/Ḍaʿīf/Mawḍūʿ* grades. Evaluation against ground-truth gradings from the *Kutub as-Sitta* (the Six Canonical Hadith Collections; i.e. Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmidhī, Sunan al-Nasāʾīand Sunan Ibn Mājah) uses accuracy, F1, RMSE, and Cohen's Kappa. The framework addresses the research gap of end-to-end automated trust scoring, providing an explainable, reproducible decision-support tool for Ḥadith scholars.

Keywords: Hadith authentication, Ilm ar-Rijal, Jarḥ wa Taʿdīl, Arabic NER, Knowledge Graph, Graph Neural Networks, semantic sentiment quantisation, multi-modal framework

ABSTRAK

RANGKA KERJA MULTIMOD UNTUK MENGUANTIFIKASI KEBOLEHPERCAYAAN PERAWI DALAM ILMU RIJAL MENGGUNAKAN ANALISIS SENTIMEN SEMANTIK DAN RANGKAIAN NEURAL GRAF

Pengesahan riwayat-riwayat Nabi Muhammad (hadis) melalui ilmu kritik perawi (ilmu rijal/al-jarh wa al-ta'dil) merupakan suatu tunjang kesarjanaan Islam, namun era digital telah mendahului kaedah-kaedah pengesahan manual. Projek ini mencadangkan rangka kerja multimodal yang mengautomasikan penskoran kebolehpercayaan perawi dengan mengintegrasikan tiga modaliti pengiraan: Pengecaman Entiti Nama (NER) Bahasa Arab dengan menggunakan model AraBERT v2 yang ditala halus untuk mengekstrak nama perawi, kunyah dan nisbah daripada teks biografi klasik (Taqrīb al-Tahḏīb karya Ibn Ḥajar al-Asqalānī dan Tahḏīb al-Kamāl karya al-Mizzī) serta rangkaian isnad (AR-Sanad 280K); Graf Pengetahuan (KG) yang dibina dengan NetworkX (~1,500 nod perawi, tepi lisan seperti heard_from dan reported_from, atribut kaya seperti tabaqa, sejarah perjalanan, dan label Jarḥ wa Taʿdīl klasik) untuk membolehkan pemeriksaan integriti struktur dan temporal; dan kuantifikasi sentimen semantik yang memetakan istilah kualitatif (contohnya, Thiqa, Ṣadūq, Kaḏḏāb) kepada pemberat kebolehpercayaan berterusan (0.00–1.00) serta menggunakan pengubah suai tepi (contohnya, 0.80 untuk tadlīs). Skor rantai akhir dihitung sebagai hasil darab pemberat nod dan pengubah suai tepi, kemudian dipetakan kepada gred Ṣaḥīḥ/Ḥasan/Ḍaʿīf/Mawḍūʿ. Penilaian terhadap penggredan kebenaran daripada Kutub as-Sittah (enam koleksi Hadis kanonik iaitu Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmidhī, Sunan al-Nasāʾī dan Sunan Ibn Mājah) menggunakan ketepatan, F1, RMSE, dan Kappa Cohen. Rangka kerja ini menangani jurang penyelidikan dalam penskoran kebolehpercayaan automatik dari hujung ke hujung, langsung menyediakan alat sokongan keputusan yang boleh dijelaskan dan dihasilkan semula untuk ulama hadis.

Kata kunci: Pengesahan Hadis, Ilmu Rijal, Al-Jarh wa al-Ta'dil, NER Bahasa Arab, Graf Pengetahuan, Rangkaian Neural Graf, kuantifikasi sentimen semantik, rangka kerja multimodal

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CHAPTER 1

INTRODUCTION

1.1 Overview

The authentication of prophetic traditions (Sunnah) through the reports attributed to the Prophet Muhammad (Hadith) forms one of the most intricate scholarly endeavours in classical Islamic civilisation. To achieve such rigorous requirements, the science of evaluating the men and women who form the chains of narration (*isnād*) through which a Hadith is transmitted is then employed. In Arabic, this science is known as *'Ilm ar-Rijal*. Additionally, an inner layer of the science known as the discipline of *Jarh wa Ta'dīl* (literally "criticism and commendation") provides the epistemological framework by which scholars assess the moral probity (*'adālah*) and mnemonic precision (*dabt*) of every individual narrator in a given chain (Kamali, 2014). Over the centuries, this civilisational practice produced vast historical corpora of biographical literature, such as Ibn Hajar al-'Asqalāni's twelve-volume Tahdhīb al-Tahdhīb, representing thousands of qualitative evaluative judgements that collectively determine the trustworthiness of the prophetic record.

The digitisation of classical texts through global repositories in recent years has made the Hadith corpus accessible on an unprecedented scale. Yet paradoxically, this very accessibility has created a new challenge; the capacity of modern scholarship to rigorously authenticate and verify Hadith at the speed of digital dissemination has not kept pace with the volume of their circulation. Recent work by Mosa (2023) highlights a pervasive issue of general anxiety, wherein the rapidity of digital access frequently surpasses the capacity to authenticate transmission chains. The manual cross-referencing labour of biographical entries across hundreds of classical volumes remains an expert-dependent process that is resistant to scale, thus presenting a compounding challenge for the field as a whole (Mosa, 2023).

In hopes of closing such gaps, this project proposes to develop a multi-modal and data-driven framework that automates and hopefully streamlines the whole process of narrator reliability evaluation while being grounded in the classical principles of Jarh wa Ta'dīl. The framework shall also aim to integrate these three advanced computational techniques:

1. Transformer-based Arabic Named Entity Recognition (NER) for the precise extraction and disambiguation of narrator identities from classical texts (Muther et al., 2023);
2. Graph Neural Networks (GNN) for modelling the structural and temporal integrity of *isnād* transmission chains (Mahmoud et al., 2024); and
3. Semantic Sentiment Quantisation to translate qualitative scholarly evaluations into a standardised, continuous reliability index (Ghriss, 2025).

The integration of these methodologies is significantly informed by foundational studies, including Mahmoud et al. (2022) on narrator disambiguation, Farooqi et al. (2024) on structured transmission datasets, and sequence-based *isnād* modelling by Najeeb (2022) and Oumer (2025). This research seeks to develop a reproducible, explicable, and epistemologically robust system that directly enhances the traditional discipline of *'Ilm ar-Rijal* through contemporary data science techniques.

1.2 Problem Background

The evaluation of Hadith authenticity through *'Ilm ar-Rijal* remains a cornerstone of Islamic scholarship, as the credibility of a prophetic tradition is fundamentally tied to the reliability of its chain of transmission (*isnād*). This rigorous verification system necessitates that every link in the chain be scrutinised for both moral integrity (*ʿadāla*) and mnemonic precision (*dabt*) (Kamali, 2022). Traditionally, this process is a labour-intensive endeavour that requires scholars to manually cross-reference thousands of qualitative biographical entries across vast historical corpuses, such as the 12-volume Tahdhib at-Tahdhib of Ibn Hajar or the foundational Andalusian Maliki assessments in Ibn ʿAbd al-Barr's At-Tamhid (Al-Asqalani, d. 852 AH; Ibn ʿAbd al-Barr, d. 463 AH).

Feasibility studies have confirmed that machine learning can indeed assist in automating *isnād* analysis, provided that the nuances of classical Arabic and the methodological standards of traditional jurisprudence are preserved (Usman, 2025; Mahmoud et al., 2022). The consonantal script and orthographic ambiguity of classical texts, however, make this particularly challenging, as minor variations in name orthography can lead to misidentification of narrators across manuscript traditions (El Ganadi et al., 2025). This results in an urgent need for Natural Language Processing (NLP) techniques to extract and disambiguate narrator identities, and for Graph Neural Networks (GNN) to model the complex relational networks those narrators form (Farooqi et al., 2024; Mahmoud et al., 2022).

Recent work has confirmed the technical viability of this approach, where transformer-based models like AraBERT have achieved high accuracy in identifying ambiguous narrators from biographical text, while GNN frameworks can generate node embeddings to capture the relative position and credibility of a narrator within the broader transmission network (Ahmad et al., 2025; Mahmoud et al., 2022). By transitioning to a systematically quantified framework, researchers can achieve higher consistency and speed in evaluation without compromising the epistemological roots of the science (Tarmom, 2022).

1.3 Problem Statements

Two primary and significant challenges remain in the computational study of Hadith despite promising advancements in Arabic NLP:

1. While current models have improved in Named Entity Recognition (NER) for extracting narrator names and lineages, there is a critical lack of computational frameworks capable of quantifying narrator reliability. Identifying a narrator is insufficient, as the framework must be able to "judge" their standing based on historical evidence (Ahmad et al., 2025).
2. Recent work by Mosa (2025) and Oumer (2025) has shown that transformer models are capable of extracting narration chains from hadith text as graphs. Nonetheless, those approaches stop at the graph, as they did not assign a reliability score to any narrator or use any actual *Jarh wa Ta'dil* praise or criticism terms to judge narrator trustworthiness. This research goes beyond graph construction by adding a quantitative reliability index and GNN-based structural analysis.

1.4 Project Objectives

The main objectives of this project are as follows,

1. To develop an NLP classification model that maps qualitative scholarly judgments to a standardised, quantitative reliability index (0.0 - 1.0).
2. To construct a Knowledge Graph representing teacher-student relationships and utilise GNNs supported with geospatial data to evaluate the structural integrity and geographical feasibility of transmission chains.
3. To evaluate the framework's accuracy by comparing automated reliability scores against established scholarly Hadith gradings (*Sahih*, *Hasan*, *Dha'if*).

1.5 Project Scope

To ensure that the research remains focused, this project operates within a clearly defined set of boundaries, a technically feasible methodology, and is evaluable within the constraints of a single academic year.

1.5.1 Data Sources

The framework will be trained and tested exclusively on digitised classical Arabic biographical texts available through the OpenITI and Shamela repositories, specifically the Taqrīb al-Tahdhīb of Ibn Hajar al-Asqalāni and the Tahdhīb al-Kamāl of al-Mizzī. These texts contain the most comprehensive and widely referenced collections of narrator biographical assessments and thus represent the most suitable ground-truth source for *Jarh wa Ta'dīl* labels.

1.5.2 Scale and Dataset

The initial experimental scope targets approximately 1,500 narrator biographical entries for building the reliability model and 1,000 validated Hadith *isnāds* drawn from the *Al-Kutub al-Sittah* (The Six Canonical Books, i.e. Sahih al-Bukhari, Sahih Muslim, Sunan Abu Dawud, Sunan at-Tirmidhi, Sunan an-Nasa'i, and Sunan ibn Majah) for end-to-end testing. While this is a small scale compared to the hundreds of thousands of available data points, this scale remains consistent with comparable feasibility studies in the literature (Mahmoud et al., 2022; Najeeb, 2022) and is sufficiently large to support statistically meaningful evaluation without exhausting available computational resources.

1.5.3 Technical Scope

The system will focus on three interconnected tasks,

1. Narrator extraction: Identifying proper names, kunya (honorifics), and nisba (place-based identifiers) from *isnād*;
2. Entity disambiguation: Resolving alias variations to canonical narrator identities via a Knowledge Graph; and
3. Reliability scoring: Mapping aggregated *Jarh wa Ta'dīl* labels to a continuous index $W = [0.0 - 1.0]$. The primary programming language is Python, utilising Arabic-specific NLP libraries (e.g., CAMeL Tools, PyArabic) and transformer models (e.g., AraBERT, MARBERT).
4. Geospatial Mapping: Extracting narrator locations (*nisba*) and mapping them to coordinates to verify if a student and teacher were in the same region at the same time.

1.5.4 Application Domain Limitations

This research is strictly concerned with the **structural evaluation of *isnād*** (*sanad* criticism). The analysis of Hadith textual content (*Matn* criticism) requires comparative jurisprudential reasoning across the four Sunni schools and falls outside the scope of this project. Furthermore, any automated reliability score produced by the framework is intended as a **decision-support tool** for scholars, not as a definitive Islamic ruling or fatwa. The epistemological authority of qualified human *muhaddithun* is expressly preserved.

1.5.5 Out of Scope

The following are explicitly excluded from this project:

- *Matn* (textual body) analysis or content-based Hadith grading.
- Coverage of Shī'a *isnād* evaluation methodologies.
- Real-time or web-based deployment of the scoring pipeline.
- Languages other than Arabic (e.g. Urdu or Persian Hadith commentaries).

CHAPTER 2

LITERATURE REVIEW

2.1 Hadith and its Structure

According to Almaany (n.d.), a Hadith (حَدِيث) is an Arabic noun that means "report; account; tale; narrative; yarn" or "talking; discourse; words; utterance; address; conversation; dialogue; saying; speaking; speech; talk". Islamically, hadith due to its nature as the embodiment of the direct speech, talk, words, actions, approvals, and attributes from the Prophet Muhammad himself, plays a very important role in reconstructing the Sunna ("practice") as practiced by the Prophet and his companions (Shaffat, 2024). The Sunna on the other hand is essential in shaping the human understanding behind the divinely revealed law of the Sharia. If the Quran provides the broad legal axioms, the Sunna provides the operational parameters, and the Hadith serves as the transmission medium for those parameters, consequently making the hadith corpora as the primary empirical data through which the Sunna is accessed and reconstructed (Nyazee, 2019).

In terms of structure, a hadith consists of a *sanad* (chain of transmission, literally "support", plural: *isnād*) and a *matn* (content) (Shaffat, 2024). Below is an example of a hadith and the aforementioned structure:

وَحَدَّثَنِي عَنْ مَالِكٍ، عَنِ ابْنِ شِهَابٍ، عَنْ عَلِيِّ بْنِ حُسَيْنِ بْنِ عَلِيِّ بْنِ أَبِي طَالِبٍ، أَنَّ رَسُولَ اللَّهِ صلى الله عليه وسلم قَالَ " مِنْ حُسْنِ إِسْلاَمِ الْمَرْءِ تَرْكُهُ مَا لاَ يَعْنِيهِ " .

And it was related to me (Yahya ibn Yahya al-Laythi) from Malik, from Ibn Shihab, from Ali ibn Husayn ibn Ali ibn Abi Talib, that the Messenger of Allah ﷺ said, "Part of the excellence of a man's Islam is that he leaves what does not concern him" (Mālik ibn Anas, 1989, Book 47, Hadith 3).

Based on the example hadith, the part "And it was related to me (Yahya ibn Yahya al-Laythi) from Malik, from Ibn Shihab, from Ali ibn Husayn ibn Ali ibn Abi Talib, that the Messenger of Allah ﷺ said" can be considered as the *sanad* (chain of transmission), while the quote itself, "Part of the excellence of a man's Islam is that he leaves what does not concern him" is the *matn* (content). In order to grade hadiths, the individuals specified in the *sanad* must have their reliability be verified, and this is done through the meticulous science of *'Ilm ar-Rijal*, literally the "Studies of Men" or the more common Sunni and synonymous term, *Jarh wa Ta'dil*, literally "critiscm and praise". Essentially, *'Ilm ar-Rijal* or *Jarh wa Ta'dil* functions as a metadata layer for the *sanad*, providing qualitative attributes such as reliability, memory, and lifespan for each node in the transmission graph (Indra Yuliasti et al., 2025; Ismail, 2024). A sanad therefore can be visualised as follows:

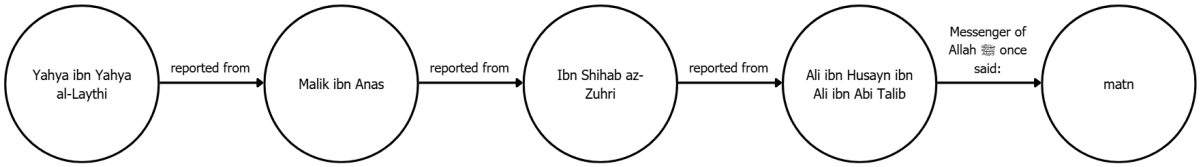


Figure 1.1: Graph Visualisation of a *Sanad*.

2.2 Impact of Hadith Grading

The grading of Hadith carries profound consequences for Islamic law (*fiqh*), theology (*'aqidah*), and especially in everyday Muslim practice. Hadith that are classified as *Sahih* (sound) or *Hasan* (good) are considered legally normative and form the second most authoritative source of Islamic jurisprudence after the Quran (Kamali, 2014). Conversely, acting upon *Da'if* (weak) or *Mawdu'* (fabricated) traditions without qualification risks introducing groundless practices into religious life (Umanah, 2024). The stakes of misclassification are therefore real and can be detrimental, as they directly affect rulings on worship, commerce, family law, and ethics observed by more than 1.9 billion Muslims worldwide.

The reliability of this grading system relies entirely on the truthfulness of the individuals who transmitted these reports across generations. A single unreliable narrator anywhere in the chain is sufficient to downgrade a Hadith, regardless of how credible the surrounding transmitters may be (Kamali, 2014). This sensitivity to individual narrator credibility implies that even small errors in evaluation, whether due to misidentification of a narrator or misapplication of a critical term, can disseminate into downstream jurisprudential decisions. Therefore, the cumulative impact of such errors, particularly in the rapidly expanding digital Hadith landscape, represents a serious epistemological risk that this research directly seeks to mitigate (Umanah, 2024; Sri Wahyuningsih, 2025).

Looking beyond just the religious domain, the science of Hadith grading has also drawn increasing attention from historians, linguists, and information scientists as one of the oldest and most systematic peer-review and source-criticism methodologies in human intellectual history (Usman, 2025). The methodological rigour of classical *muhaddithun* (Hadith scholars), who developed chains of trust, biographical verification, and cross-referencing centuries before modern academic citation standards, offers a compelling analogue to contemporary knowledge verification challenges (Ghriss, 2025). Applying computational tools to this tradition, therefore, carries interdisciplinary significance that extends well beyond Islamic studies alone (A'thoina & Hasballah, 2025).

2.3 Hadith Authentication Process

The classical process of authenticating a Hadith is a dual-track methodology that evaluates both the *sanad* (chain of narrators) and the *matn* (textual body) of a tradition (Kamali, 2014). Sanad criticism, the focus of this research, concerns the continuity and credibility of the transmission chain from its originator (typically the Prophet Muhammad ﷺ) to the collector who recorded it. Each narrator in the chain must be proven to have met the preceding narrator, to have possessed an accurate memory, and to have been morally upright throughout their life. Any break in the chain (*inqita'*) or identified moral failing in a transmitter is sufficient grounds to weaken or reject the report entirely (Ahmed & Anas, 2025).

The grading system that emerges from this process is hierarchical and multi-tiered. At the apex sits the *Sahih* ("sound") Hadith, which satisfies all criteria of chain continuity, narrator credibility, and absence of hidden defects (*'illah*) or contradictions (*shudhadh*). Below this are degrees of *Hasan* ("good"), where minor shortcomings in narrator memory are present but do not fatally undermine the report. *Da'if* ("weak") Hadiths fail one or more of these conditions, whereas *Mawdu'* ("fabricated") traditions are those for whose narrators deliberate falsehood has been established (Kamali, 2014). This taxonomy forms the classification target for the automated scoring system proposed in this research, where the aim is to computationally approximate the output of this traditional grading process (Ahmad et al., 2025).

The process of collecting the qualitative assessments that inform these grades is documented in a vast body of biographical literature known as *kutub ar-rijal* (books on narrators). Scholars such as Yahya ibn Ma'in, Ahmad ibn Hanbal, and later Ibn Hajar al-'Asqalani dedicated their careers to interviewing, cross-referencing, and documenting the reliability status of thousands of narrators (Mghari et al., 2024; A'thoina & Hasballah, 2025). These books contain the raw *Jarh wa Ta'dil* labels, i.e. terms of praise and criticism that this project aims to computationally extract, quantify, and aggregate into an automated reliability index.

2.4 'Ilm ar-Rijal: An Expansive Biographical Study

'Ilm ar-Rijal ("the science of narrators") or also synonymously known as *Jarh wa Ta'dil* ("criticsm and praise") is one of the most elaborate biographical knowledge systems ever developed in any civilisation (Ahmed & Anas, 2025). At its peak, classical scholars tracked the life histories, teachers, students, travel records, and credibility assessments of over 500,000 narrators, a feat of multi-generational data collection that has no direct parallel in pre-modern scholarship (Usman, 2025). The core output of this science, the biographical dictionary (*mu'jam* or *tabaqat*), catalogues narrators by generation, geographic region, or alphabetical order, and assigns each an evaluative standing based on the consensus of prior critics (Maraoui et al., 2022).

The terminology employed within *'Ilm ar-Rijal* is highly technical and hierarchically structured. Terms of commendation (*ta'dil*) range from the superlative *Thiqah Thiqah* (supremely reliable) down to *Saduq* (truthful but occasionally erring), with each grade carrying different implications for the admissibility of a narrator's reports. Terms of criticism (*jarh*) range from mild reservations such as *Layyin* ("soft", i.e., slightly weak) to severe condemnations such as *Kadhdhab* ("liar") or even *Dajjal* ("arch-deceiver") (Kamali, 2014). The precise semantic weight of each term and the conditions under which it strengthens or overrides competing assessments are the subject of a dedicated sub-discipline (Mghari et al., 2024), one that this project attempts to operationalise through Semantic Sentiment Quantisation.

A central computational challenge in this science is the problem of resolving **narrator identity**. The Arabic naming convention comprises a given name (ism), teknonym (*kunya*), patronym, matronym or a series of it (*nasab*), tribal affiliation (*nisba*), and nickname (*laqab*), which means that a single narrator may be referred to by dozens of orthographic variants across different texts and centuries (Mahmoud et al., 2022). The scholar known canonically as Abu Dawud as-Sijistani, for example, may appear across manuscripts in numerous abbreviated or dialectally inflected forms. Disambiguating these variants into a single canonical identity is a prerequisite for any reliable automated grading system, and is the primary challenge that the Knowledge Graph and NER components of this research are designed to solve (Najeeb, 2022; Mosa, 2025).

2.5 Advancements in Arabic Named Entity Recognition (NER)

Modern processing of classical Arabic text has moved decisively away from rule-based constraints toward deep bidirectional representations. Research by Mahmoud et al. (2022) using the AR-Sanad 280K dataset established a new baseline for narrator disambiguation, proving that large-scale datasets are essential for handling the morphological density of biographical entries.

The current "State of the Art" (SOTA) by Mosa (2025) and Oumer (2025) utilises Pretrained Language Models (PLMs). Alghamdi et al. (2025) demonstrated that Transformer-based models like AraBERT and MARBERT significantly outperform previous iterations in identifying fraudulent content and extracting technical entities. These advancements facilitate high-accuracy Entity Linking, allowing the system to resolve identities across centuries even when names are identical (Mahmoud et al., 2022; Najeeb, 2022).

2.6 Knowledge Graphs (KG) for Entity Disambiguation

A Knowledge Graph (KG) is a structured representation of real-world entities and the semantic relationships between them, encoded as a directed graph of nodes and edges (Ji et al., 2022). In the context of *'Ilm ar-Rijal*, this structure is particularly well-suited to the problem of narrator identity resolution: each node represents a canonical narrator identity, edges encode transmission relationships (teacher-student pairs), and node attributes store the accumulated biographical metadata, including *Jarh wa Ta'dil* assessments drawn from classical *kutub ar-rijal* (Maraoui et al., 2022; Shafie, 2021). Unlike flat relational databases, KGs allow queries that traverse multi-hop relationships, enabling the system to infer that two orthographically different name mentions in separate manuscripts refer to the same historical individual based on their shared network neighbourhood (Hogan et al., 2021).

Within the domain of digital Hadith studies, several prominent Knowledge Graphs (KGs) have been developed to model the complex interdependencies of *isnād* chains (Alam, 2021). Systems such as **NarratorsKG** encode the topological structure of narrator transmission networks, mapping tens of thousands of distinct identities alongside their diverse *kunya* and *nisba* variants (Mahmoud et al., 2024). In a typical hybrid retrieval pipeline, an NER model extracts name mentions, which are then resolved against the KG's candidate pool using both local semantic context and global graph signals (Mahmoud et al., 2024). Specifically, graph-level metrics like **PageRank centrality** are utilised to weigh a narrator's historical prominence; this helps resolve ambiguities where purely lexical matching fails, a common issue in the classical corpus where a single name may refer to five or more distinct individuals (Mosa, 2025).

More recently, the integration of KGs with Transformer-based models has defined the current state of the art. Mosa (2025) proposed a KG-Transformer framework that encodes narrator node embeddings directly into the attention mechanism of a pre-trained Arabic language model, allowing the model to simultaneously process the local textual context of a name mention and its global relational position within the transmission network. This synergy between structural and semantic representations addresses what earlier approaches treated as separate problems, which are text understanding and graph reasoning, and produces entity disambiguation accuracy that approaches or exceeds 97% on the AR-Sanad 280K-v2 benchmark (Mahmoud et al., 2024; Mosa, 2025). The construction and querying of such a KG therefore constitute a foundational prerequisite for the GNN-based reliability analysis described in the following section.

2.7 Advancements in Graph Neural Networks (GNN) for *Isnād* Analysis

Traditionally, the "structural integrity" of an *isnād* was analysed using Social Network Analysis (SNA), focusing on explicit metrics like degree centrality and betweenness to visualise transmission flows (Shafie, 2026; Siddiqui et al., 2023). While useful for visualisation, SNA is limited in its ability to learn latent relational features (Apgar et al., 2023).

The field has recently evolved toward Graph Neural Networks (GNN) and Knowledge Graphs (KG). Mahmoud et al. (2024) introduced narrator identification by utilising graph query embeddings and a knowledge graph (NarratorsKG), while Mosa (2025) proposed a Knowledge Graph-Transformer framework to synergise structure and semantics. Unlike SNA, GNNs can represent a narrator's "reliability" as a latent feature influenced by their entire neighbourhood of teachers and students. This allows for the detection of "hidden" patterns, such as temporal inconsistencies where a student claims to have heard from a teacher they could not have met, thus automating a core task of the *muhaddithun*.

A recent master's thesis (Oumer, 2025) successfully fine-tuned AraT5 and CAMeLBERT-CA to turn hadith text into relational tuples and graph representations. While it is a useful structural work, it explicitly leaves out narrator reliability scoring, entity linking, and any quantification of *Jarh wa Ta'dil* terms. This project aims to take graphs from the work as a starting point and add the missing "trust layer", something clearly stated as out of scope in the aforementioned thesis.

2.8 Advancements in Semantic Sentiment Quantisation

A major challenge in automating Hadith grading is translating qualitative scholarly praise (*Ta'dil*) or criticism (*Jarh*) into numerical formats. Recent feasibility studies by Ahmad et al. (2024) confirm that Machine Learning is increasingly capable of automating *isnād* analysis when technical labels are correctly weighted.

Current advancements focus on Semantic Sentiment Quantisation, where technical terms such as *Thiqah* (reliable), *Saduq* (truthful), and *Munkar* (rejected) are mapped to specific reliability weights. Mghari et al. (2024) introduced Narrator2Vec, which uses word embeddings to create dense representations of a narrator's standing. This allows for a continuous reliability index $W \in [0, 1]$, where the framework calculates a composite probability score for an entire chain by aggregating the individual quantised weights of each narrator. This move toward "Explainable AI" ensures that the quantitative output remains rooted in traditional Islamic epistemology (I'thoina & Hasballah, 2024). The synthesis of these three advancements, i.e. transformer-based identity resolution, GNN-driven structural analysis, and sentiment-based weight quantisation, forms the multi-modal foundation of this research.

2.9 The Research Gap: Automated Trust & Reliability Scoring

Despite the significant advancements documented in the preceding sections, a critical gap persists at the intersection of these research streams: no published study has yet delivered a complete, end-to-end pipeline that unifies narrator extraction, identity disambiguation, and probabilistic reliability scoring into a single, evaluatable system. The majority of existing work ceases at either the extraction or disambiguation stage, treating the production of a quantified trustworthiness score as a future direction rather than an achieved outcome (Ahmad et al., 2025). While text-based automated classification methods do exist (Tarmom,

2022), what remains absent is an end-to-end system that unifies knowledge graphs with probabilistic reliability scoring.

The separation between textual NLP work and graph-structural work further compounds this gap. Studies employing AraBERT or similar transformer models for NER do not, as a rule, feed their outputs into Knowledge Graph or GNN frameworks for structural validation (Alghamdi et al., 2025). Conversely, graph-based analyses of *isnād* networks rarely incorporate the semantic richness of biographical *Jarh wa Ta'dil* texts as node features (Mosa, 2025; Mahmoud et al., 2024). This disciplinary fragmentation means that existing systems capture either the semantic dimension of narrator reliability or the structural dimension, but not both simultaneously. The multi-modal integration that would make automated grading genuinely robust remains a largely unexplored frontier (Ghriss, 2025).

The standardisation of evaluation metrics also presents a significant gap. While individual studies report precision, recall, or F1-scores for their specific sub-tasks, no shared benchmark protocol exists for assessing the accuracy of an end-to-end automated grading system against the ground-truth classifications found in the *Kutub as-Sitta* (Farooqi et al., 2024). This absence of a common evaluation framework makes it difficult to compare approaches across studies or to claim generalised progress toward the broader goal. Establishing such a benchmark protocol is therefore a secondary but important contribution of the present research (Mahmoud et al., 2022).

The closest existing work by Oumer (2025) builds a pipeline from hadith text to a graph of narrators. While it is a real contribution, it does not answer the question that ʻIlm ar-Rijal actually asks: given this chain of narrators, how reliable are they? The output in the aforementioned work is a graph, while the aimed output of this project is a graded Hadith (Sahih/Hasan/Daʻif) with an explainable score. No published study has yet combined semantic sentiment from *Jarh wa Taʻdil* texts with GNN-based structural analysis to produce a narrator reliability score.

This project directly targets these gaps by proposing a unified pipeline that integrates transformer-based NER, Knowledge Graph entity linking, GNN structural analysis, and Semantic Sentiment Quantisation into a single framework evaluated against pre-graded Hadith from the canonical collections. In doing so, it advances the field from the current state of fragmented sub-system studies toward a reproducible, multi-modal solution grounded in both classical Islamic epistemology and contemporary machine learning practice.

CHAPTER 3

PROJECT METHODOLOGY

3.1 Research Design

This project adopts an experimental research design underpinned by the **Design Science Research Methodology** (DSRM). The central artefact is an end-to-end automated pipeline that ingests classical Arabic biographical text, extracts and disambiguates narrator entities, and produces a continuous reliability score for each narrator, thereby operationalising the traditional science of *Jarh wa Ta'dīl* through computational means. Development proceeds through three sequential phases, each corresponding to a distinct research objective defined in Section 1.4. Each phase produces a testable intermediate output that feeds into the next, ensuring that failures can be isolated and addressed incrementally. The methodology is structured as follows:

Phase A: Establishes the data foundation;

Phase B: Constructs the extraction and disambiguation engine; and

Phase C: Implements the scoring model and end-to-end evaluation.

This framework will add a probabilistic trust score for each narrator and use graph neural networks to evaluate chain-level reliability, defining the novelty of this project. Together, these phases form the technical pipeline illustrated in Section 3.5.

3.2 Phase A: Data Ingestion and Preprocessing

3.2.1 Data Sources

The primary data sources are twofold:

1. Biographical Texts: Digitised editions of Taqrīb al-Tahdhīb (Ibn Hajar al-'Asqalānī) and Tahdhīb al-Kamāl (al-Mizzī), sourced from the OpenITI corpus and the Shamela Digital Library. These texts provide the *Jarh wa Ta'dīl* labels (qualitative scholarly assessments) that form the ground-truth input for the reliability scoring model.
2. Benchmark *Isnād* Dataset: The AR-Sanad 280K dataset, which contains approximately 280,000 synthetically augmented *isnād* chains with annotated narrator boundaries used for training and evaluating the NER component.

3.2.2 Preprocessing Pipeline

Classical Arabic texts present unique preprocessing challenges due to their consonantal script, optional diacritical marks (*tashkīl*), and the absence of standardised punctuation in

historical manuscripts. The preprocessing pipeline addresses these challenges through the following steps:

1. Encoding Normalisation: All source texts are converted to UTF-8, with Byte Order Marks (BOM) and legacy Windows-1256 artefacts stripped.
2. Unicode Normalisation: The Arabic text is normalised to NFKC form. Variant forms of Alif (آ ,إ ,أ) are collapsed to bare Alif (ا), and Ta Marbuta (ة) is normalised to Ha (ه) where contextually appropriate, using the CAMeL Tools normalisation utilities.
3. Diacritic Stripping: Optional *tashkīl* (short vowel marks) and *shadda* are removed to produce a consonantal skeleton, reducing orthographic variance without sacrificing semantic content.
4. Tokenisation: Sentences are tokenised using the Farasa segmenter (Mahmoud et al., 2022), which handles Arabic clitics (conjunctions, prepositions, and definite articles prefixed to words) by splitting them into separate tokens. This step is critical for downstream NER, as narrator names frequently appear with attached particles (e.g., وعن → و + عن, literally "and" + "from" -> "and from").
5. *Isnād–Matn* Segmentation: Each Hadith record is split into its *isnād* (chain) and matn (body) segments using transmission keyword detection. Transition markers such as حدثنا ("narrated to us"), أخبرنا ("informed us"), and عن ("from") are used as boundary signals. Only the *isnād* portion is retained for downstream processing.

3.2.3 Expected Output

The output of Phase A is a cleaned, tokenised corpus of approximately 1,500 biographical entries and 1,000 segmented *isnād* chains, stored in a structured JSON format with the following schema per record: `{id, raw_text, normalised_text, tokens[], source_book, entry_type}`.

3.3 Phase B: Named Entity Recognition and Knowledge Graph Construction

3.3.1 NER Model Architecture

Narrator extraction is performed using a fine-tuned Transformer-based model. The base model is **AraBERT v2**, a pre-trained bidirectional encoder trained on a large Modern Standard Arabic corpus (Mahmoud et al., 2022). The model is fine-tuned on the AR-Sanad 280K dataset for a token classification task using the BIO (Beginning, Inside, Outside) tagging scheme. The following entity types are annotated:

Table 1: Example of Narrator Entity for the Companion Abu Hurayra

Tag	Entity Type	Example (Arabic)	Transliteration
B-NARRATOR	Beginning of narrator name	عبد ٱلرحمن	ʿAbd ar-Raḥmān
I-NARRATOR	Inside narrator name	بن صخر	bin Ṣakhr
B-KUNYA	Beginning of the teknonymic honourific	أبو	Abū
I-KUNYA	Inside the teknonymic honourific	هريرة	Hurayra
B-NISBA	Beginning of geographic/tribal affiliation	الدوسي	Ad-Dawsī
I-NISBA	Inside geographic/tribal affiliation	الزهراني	Az-Zahrānī
O	Outside any entity	حدثنا	"We narrated"

The model is trained for 10 epochs with a learning rate of 2^{e-5}, using the AdamW optimiser with a linear warmup schedule over the first 10% of training steps. Evaluation is performed on a held-out 20% test split of the AR-Sanad dataset, with **micro-averaged F1-score** as the primary metric.

3.3.2 Knowledge Graph Construction

Extracted entity mentions are linked to canonical narrator identities stored in a **Knowledge Graph (KG)**. The KG is constructed using the Python 'NetworkX' library (with optional migration to Neo4j for scalability) and populated from the biographical entries in the Tahdhīb al-Kamāl. Each node in the KG represents a unique narrator and stores the following attributes:

Table 2: Attributes in a Rich Narrator Node

Node Attribute	**Description/Purpose**
ID	For entity disambiguation.
Name	The canonical name of the narrator.
Nasab	Patronym or chain of lineage.
Kunya	Teknonym (e.g., Abu Muhammad).
Nisba	Attributions based on tribe, place, or trade.
Common Name	The most famous or commonly used name when referring to the entity. May help entity disambiguation.
Birth/Death Dates	Dates for temporal integrity checks (where available).
Tabaqa	Generational marker to prevent anachronisms.
Ethnic/Legal Status	Helps determine if the transmission is localised/regional or widely accepted, and aids entity disambiguation.
Teacher/Student relationships (with duration)	Acts as the directed edges between nodes.
Travelled locations	To detect any inconsistencies or contradictions.
Recension (where available)	Boosts reliability if known to produce a critical revision of a document, such as the *Muwatta' Malik*.
Reliability	*Jarh wa Ta'dīl* labels are divided into inferred (machine-made conclusion) and classical (ground label truth).



Figure 2.1: Example of a Rich Narrator Node

Edges represent **transmission relationships** (teacher → student) extracted from *isnād* chains. When the NER model produces a name mention, the KG is queried using a combination of string similarity (Levenshtein distance on the consonantal skeleton) and graph neighbourhood matching (shared teachers/students) to resolve the mention to its most probable canonical identity.

To prevent any anachronisms and further augment the reliability of this system, an extended metadata system of *Tabaqat* (generations) based on Ibn Hajar's 12-tiered generational system will be implemented on every narrator. The generational system will cover from 622 CE (starting of the Islamic calendar) until 870 CE (around the mid-3rd century AH), and the narrators would be mapped to these discrete categories:

1: The Sahaba (Companions of the Prophet).
2: Senior *Tabi'un* (Followers of the Companions).
3: Intermediate *Tabi'un* (Followers of the Followers).
4–6: The Junior *Tabi'un*.
5-6: Immediate students (Successors) of the Junior *Tabi'un*.
7–9: The "Great Successors."
10–12: The students of the students, ending around the mid-3rd century AH.

3.3.3 Expected Output

The output of Phase B is a populated Knowledge Graph containing approximately 1,500 canonical narrator nodes with linked aliases, biographical metadata, and transmission edges, alongside an NER model capable of extracting narrator mentions from unseen *isnād* with a target F1-score of ≥ 0.90.

3.3.4 Enriched *Sanad* Knowledge Graph

Each narrator node stores the full set of attributes listed in Table 1.2 (Section 3.3.2), including birth/death dates, travel history, recension, and inferred reliability score. For visual clarity, only ID, common name, tabaqa, and reliability are shown here.

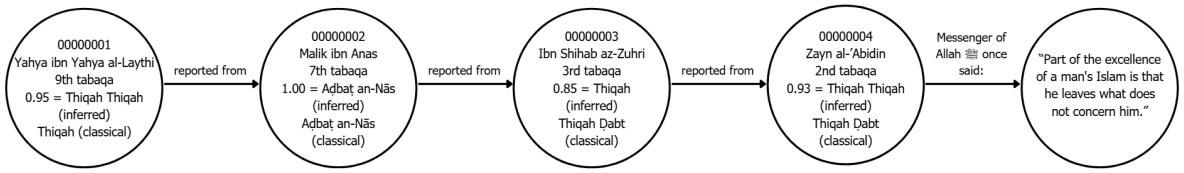


Figure 2.2: Enriched *Sanad* Knowledge Graph Mock-up

3.4 Phase C: Probabilistic Trust Scoring and Evaluation

3.4.1 Continuous Reliability Scoring

The core innovation of this phase is the translation of qualitative *Jarh wa Ta'dīl* labels into a continuous reliability score on the interval [0.0, 1.0], initially developed by Ghriss (2025) in their Iterative Co-Update Algorithm (ICUA) framework. This framework specifically formalises narrator reliability scores (R_N) on a scale of 0 to 1. This is achieved through **Continuous Reliability Scoring**, which proceeds in two steps:

1. Term Mapping: Each classical evaluative term is assigned a base weight according to the hierarchical grading scale established in the classical Hadith sciences. For example:

Table 3: Example of Term Weighting

Term (Arabic)	Transliteration	Grade	Base Weight
أضبط الناس	*Aḍbaṭ an-nās*	Select highest *ta'dil*	1.00
ثقة ثقة	*Thiqa thiqa*	Highest *ta'dil*	0.95
ثقة ضابط	*Thiqa ḍābiṭ*	Trustworthy and precise	0.90
ثقة	*Thiqa*	Strong *ta'dil*	0.85
صدوق	*Ṣadūq*	Moderate *ta'dil*	0.70
ليّن	*Layyin*	Mild *jarh*	0.20
كذّاب	*Kaḏḏāb*	Severe *jarh*	0.00

2. Aggregation: Where multiple scholars have assessed the same narrator using different terms, the individual weights are aggregated into a single composite score. Two aggregation strategies will be compared:
    a. A **weighted arithmetic mean**, where the weight of each assessment is influenced by the historical prominence of the assessing scholar; and
    b. a **Bayesian posterior estimation,** where each successive assessment updates a prior distribution over the narrator's reliability.

Both aggregation strategies will be implemented and evaluated on a validation set of 200 hadiths with known classical gradings. The strategy that yields a higher Cohen's Kappa agreement with the ground-truth grades will be selected for the final pipeline. Preliminary experiments (Ghriss, 2025) suggest that Bayesian posterior estimation is more robust when multiple conflicting Jarh wa Ta'dil assessments exist, but the weighted mean is simpler and less prone to overfitting. Therefore, both will be compared empirically.

3.4.2 Chain-Level Scoring

Once each narrator in a given *isnād* has been assigned an individual reliability score w_i, the chain-level reliability R is computed as the product of individual scores:

$$R(chain) = \prod w_i \text{ for all narrators } i \text{ in the chain.}$$

This multiplicative model developed by Ghriss (2025) will be employed, though with slight changes. The original model demonstrates that the score of a transmission path can be mathematically formulated using methods like the "product" or "average of reliabilities," though it ultimately leans toward a minimum "weakest link" function. The chain-level score R is then mapped to the traditional grading categories (*Sahih*, *Hasan*, *Da'if*, *Mawdu'*) using threshold boundaries calibrated against the ground-truth labels in the evaluation set. To further enhance the reliability of this model, the Edge Modifier (m_{edge_i}) is added in the multiplication model to address transmission-specific risks such as *tadlis* cases, where a narrator would say "from from (narrator)" instead of "from (narrator) from (narrator)", breaking the expected continuous entity-to-entity chaining, thus yielding the following model:

$$R(chain) = \prod w_{node_i} \times m_{edge_i}$$

where w_{node_i} is the *Jarh wa Ta'dil* weight for all narrators i in the chain, and m_{edge_i} is the edge modifier applied if there are known risk cases, with the value reducing from the default 1.00 depending on the severity of the risk cases. The following table lists the standardised edge modifier penalties when dealing with *sanad* risk cases:

Table 4: Standardised Modifier Range for Risk Cases

Risk Case	Classical Term (Transliterated)	Modifier Range	Example	Notes
Tadlis (ambiguous attribution)	*Mudallis*	0.70–0.85	Ibn Shihāb az-Zuhri	Default = 0.80 unless stronger evidence
Poor memory (due to old age)	*Ikhtalaṭa / Taghayyara ḥifẓuhu*	0.85-0.90	Sufyān aṯ-Thawrī	Higher range to not penalise reliable narrators heavily.
Shāḏ (singleton oddity)	*Shāḏ*	0.90-0.95	ʿAbd al-Wāḥid ibn Ziyād	Minor penalty due to known Shādh with Thiqah narrators.
Unknown identity	*Majhūl* (unknown)	0.30–0.50	Muḥammad ibn ʿAmr ibn Harith (not much biographical data, nor being cited by other narrators)	Base weight is already low from lack of data; thus, an extra penalty is not needed.
Accusation of lying	*Muttaham bil-kaḏib*	0.00–0.30	Muḥammad ibn Marwān as-Suddī	Already captured in base weight (should be near 0), thus not needed.

Only risk cases of tadlis, poor memory and šāḏ remain after careful considerations as described in the notes of Table 4 to avoid further penalising until 0.00, which would render the quantification goal of this project as futile.

Geospatial consistency is checked for every edge. For each narrator, the travelled locations attribute (Table 2) is geocoded using the Geopy library with a gazetteer of classical Islamic cities (e.g., Madina, Kufa, Basra, Damascus, Cordoba). If both the student and the teacher have at least one location with a known year of residence (derived from birth/death dates and travel history), the system verifies whether their lifespans and geographic movements could have enabled a meeting. If a student's known locations do not overlap with the teacher's at any plausible time, a geospatial penalty factor $g_{edge} = 0.70$ is multiplied into the edge modifier m_{edge}. If location data is missing or uncertain, no penalty is applied ($g_{edge} = 1.00$). This penalty is applied independently of the tadlis or ikhtilat modifiers.

3.4.3 Evaluation Protocol

The automated grades are evaluated against **ground-truth Hadith classifications** drawn from pre-graded collections in the *Kutub as-Sitta* (the Six Canonical Hadith Collections), i.e. Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmidhī (Sunan at-Tirmidhī), Sunan al-Nasāʾī (al-Mujtabā) and Sunan Ibn Mājah. The following metrics are reported:

Table 5: Purposes of Metrics Used in this Project

Metric	Purpose
Accuracy	Overall proportion of correctly classified Hadiths.
Precision/Recall/F1	Per-class performance for each grading category.
RMSE	Root Mean Squared Error between predicted chain scores and expected ordinal grades.
Cohen's Kappa	Inter-rater agreement between automated and classical human grades.

A baseline comparison is conducted against a naive majority-class classifier and a simple lexical lookup model (direct term-to-grade mapping without graph features) to quantify the contribution of each pipeline component. To model structural and temporal integrity, a lightweight two-layer Graph Convolutional Network (GCN) is implemented using PyTorch Geometric. Node features include the base reliability weight wnodei, tabaqa (encoded as an integer), and a binary flag indicating whether the narrator has a known geographic region. Edge features include the transmission method (explicit direct vs. ambiguous) extracted via pattern matching. The GCN predicts an edge probability for each teacher-student pair; probabilities below 0.50 trigger a manual review flag for potential anachronisms (e.g., a student claiming to have heard from a teacher who died before the student was born). The final node embeddings are also used to compute PageRank-like centrality scores, which serve as an auxiliary feature for outlier detection.

3.5 Multi-Modal Framework Flowchart

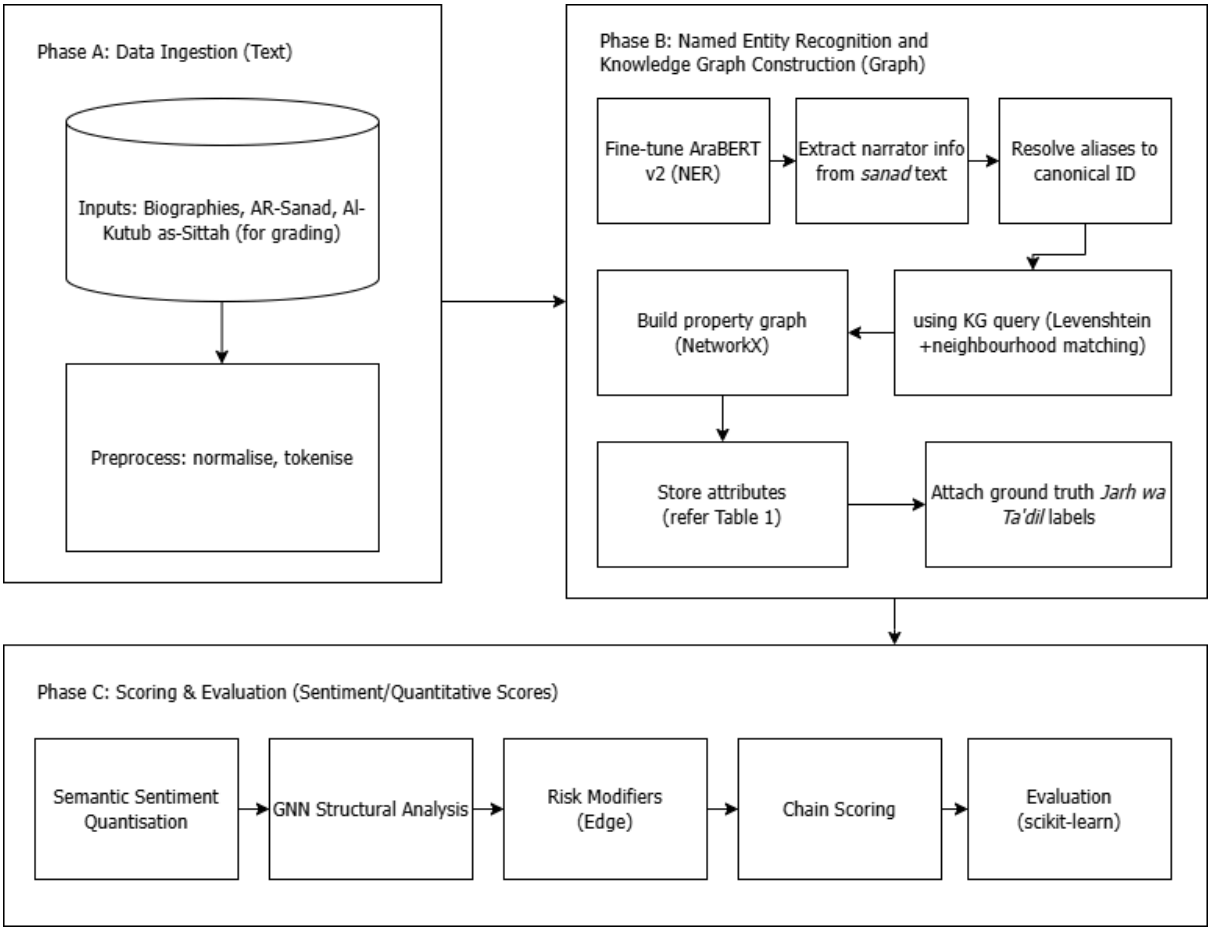


Figure 2.3: The Multi-Modal Framework Flowchart Summarising the Methodology

3.6 Flow Chart of Activities

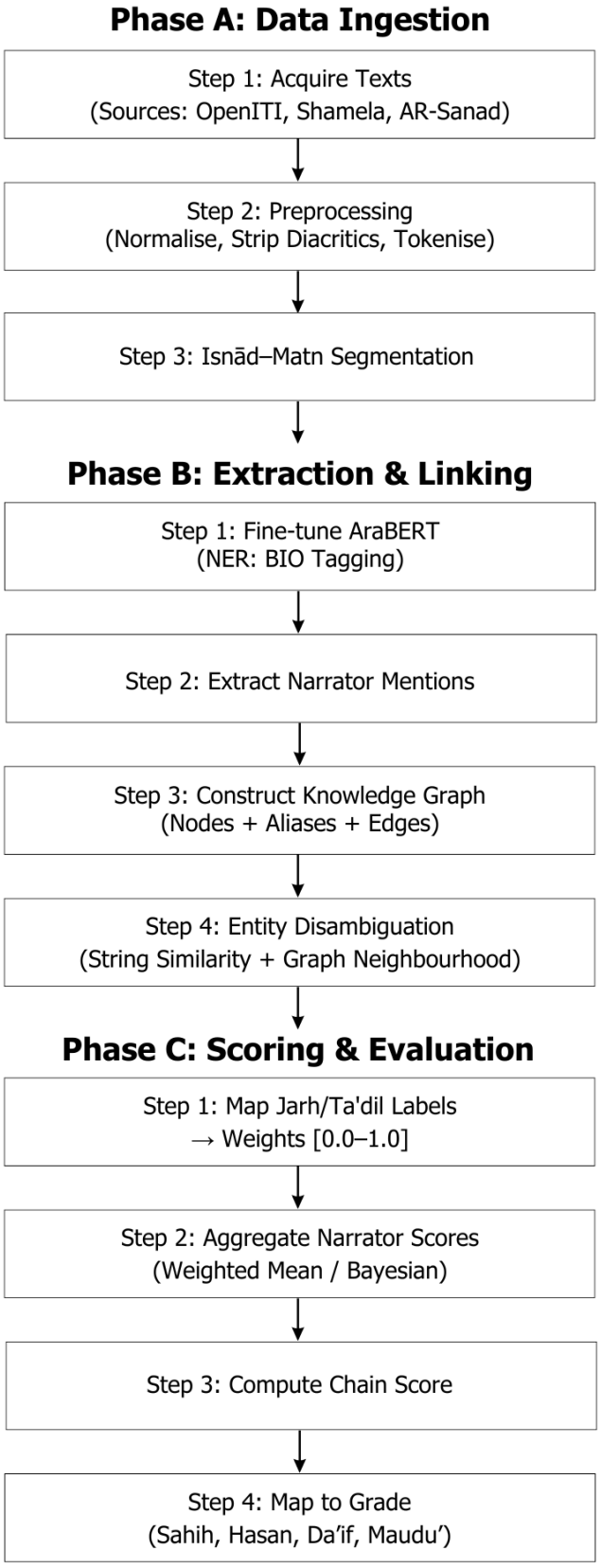


Figure 2.4: The Flowchart that Summarises the Methodology of this Project

3.7 Milestones and Dates

The following table shows the weekly milestones with their deliverable respectively:

Table 6: Milestones and Deliverables by Week

Week	Milestone	Deliverable
2-2025/2026		
1-2	Project Initialisation	Finalised project summary; supervisor approval
3-4	Literature Review	Completed Chapter 2; dataset acquisition
5-6	Methodology Design	Completed Chapter 3; proposal draft
7	Proposal Defence	Oral defence and proposal submission
8-9	Phase A Implementation	Cleaned corpus; preprocessed JSON dataset
10-11	Phase B Implementation	Trained NER model; populated Knowledge Graph
1-2026/2027		
1-2	Phase C Implementation	Trust scoring module; preliminary evaluation results
3-4	Thesis Report Preparation	Written report drafting
5-6	Thesis Presentation Preparation	Final thesis presentation drafting

3.8 Gantt Chart

Year	2026										2027	
Project Activities	M	A	M	J	J	A	S	O	N	D	J	F
Title Selection												
One-Paged Summary Submission												
Preparation of Proposal												
Proposal Submission												
Proposal Defense												
Phase A: Data Ingestion and Preprocessing												
Dataset Acquisition												
Preprocessing Pipeline												
Phase B: Named Entity Recognition and Knowledge Graph Construction												
AraBERT Fine Tuning (NER)												
Knowledge Graph Construction												
Phase C: Probabilistic Trust Scoring and Evaluation												
Sentiment Quantisation Module												
Chain Scoring & Evaluation												
Reporting												
Thesis Preparation												
Thesis Presentation Preparation												
Thesis Submission												
Thesis Presentation (VIVA)												

3.9 Tools and Technologies

Below are the tools, libraries and technologies used to build this project, along with their purpose for the project:

Table 7: Tools, Libraries, and their Purposes by Category

Category	Tool/Library	Purpose
Language	Python 3.10+	Primary development language
NLP Preprocessing	CAMeL Tools, PyArabic, Farasa	Arabic normalisation, tokenisation, segmentation
Transformer Model	AraBERT v2 (HuggingFace)	Fine-tuned NER for narrator extraction
Deep Learning	PyTorch, Transformers (HuggingFace)	Model training and inference
Graph Processing	NetworkX/Neo4j	Knowledge Graph construction and querying
Geocoding	Geopy	To convert classical Arabic place names (*nisba*) into latitude/longitude.
Mapping	Folium	To generate static maps showing the "geographical journey" of a hadith.
Evaluation	scikit-learn	Classification metrics (F1, Precision, Recall, Kappa)
Data Format	JSON, CSV	Structured intermediate data storage
Version Control	Git/GitHub	Source code and experiment tracking
Documentation	Markdown, LaTeX	Proposal and report writing

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APPENDICES